

Bile acids influence the growth, oestrogen receptor and oestrogen-regulated proteins of MCF-7 human breast cancer cells.

The effects of the major human serum bile acid, glycochenodeoxycholic acid (GCDC), as well as unconjugated chenodeoxycholic acid (CDC), on the MCF-7 human breast cancer cell line have been studied in vitro under oestrogen and bile acid deprived culture conditions. GCDC increased the growth of the breast cancer cells over the range 10-300 microM. At concentrations in excess of the bile acid binding capacity of the medium cell growth was prevented. In contrast 10 microM CDC tended to reduce cell growth. Oestrogen (ER) and progesterone (PgR) receptors, pS2 and total cathepsin D were quantified by monoclonal antibody based immunoassays. Ten to 100 microM GCDC and 10 microM CDC down-regulated ER protein and this was accompanied by induction of the oestrogen-regulated proteins PgR, pS2 and possibly cathepsin D, including increased secretion of the latter two proteins into the culture medium. All these changes were quantitatively similar to those observed with 10 nM oestradiol. The bile acid effects on ER and PgR were not due to interference with the assay procedures. Cells incubated with 50 microM GCDC or 10 microM CDC had higher pmolar concentrations of the bile acids than controls. This study suggests that naturally occurring bile acids influence the growth and steroid receptor function of human breast cancer cells.